

DAY 1: OVERVIEW OF HARDWARE COMPONENTS

Computer Hardware — Theory & Assignment

Theory (1 Hour)

Computer hardware refers to the physical parts of a computer system that can be seen and touched. The main components work together to process information, run applications, store data, display graphics, and provide power to the system.

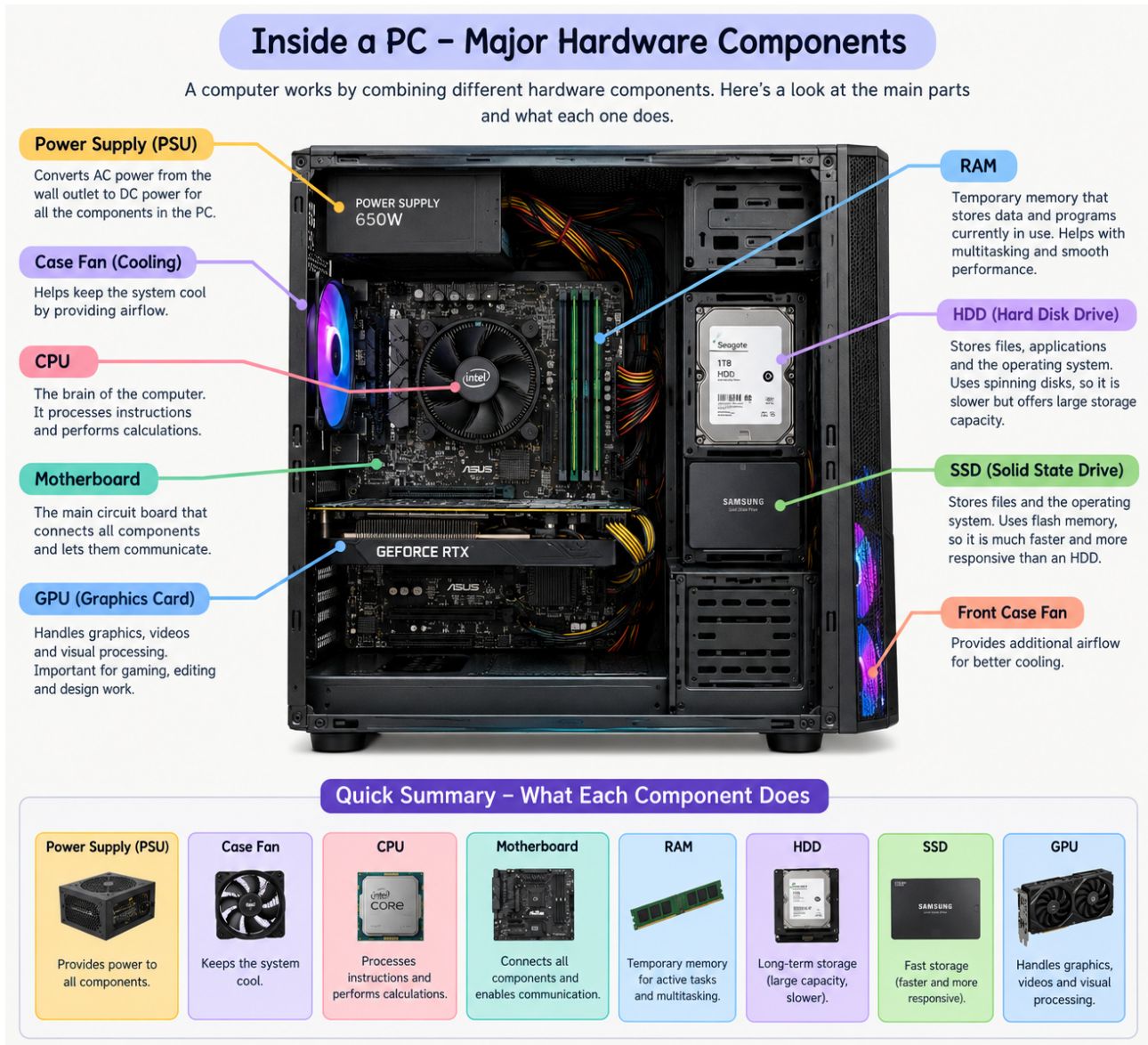
1. Motherboard	The main circuit board of a computer. It connects the CPU, RAM, GPU, storage devices, and other hardware components and allows them to communicate with each other.
2. CPU (Central Processing Unit)	The CPU is often called the brain of the computer. It processes instructions, performs calculations, and controls many of the computer's operations.
3. RAM (Random Access Memory)	RAM is temporary, high-speed memory used to hold data and programs that are currently in use. More RAM can help a computer handle multiple tasks smoothly.
4. GPU (Graphics Processing Unit)	The GPU processes images, videos, and other graphics. It is especially important for gaming, video editing, 3D design, and other graphics-intensive work.
5. Storage — HDD and SSD	Storage keeps the operating system, applications, documents, photos, and other files for long-term use. HDDs use spinning magnetic disks, while SSDs use flash memory and are generally faster.
6. Power Supply Unit (PSU)	The PSU converts electrical power from an outlet into the appropriate power needed by the computer's internal components.
7. Cooling Fans	Cooling fans move air through the computer case to help remove heat and keep components operating at suitable temperatures.

Importance of Each Component

Component	Importance
Motherboard	Connects all major hardware components and enables communication between them.
CPU	Executes instructions and performs calculations.
RAM	Provides fast temporary workspace for active programs and tasks.
GPU	Handles graphics, images, videos, and visual processing.
HDD / SSD	Provides long-term storage for the operating system, applications, and files.
PSU	Supplies the required electrical power to the computer components.
Cooling Fans	Helps prevent overheating and maintain stable performance.

Assignment: Labeled PC Hardware Diagram

The diagram below shows the major hardware components of a desktop PC and their functions.



Conclusion: A computer works as an integrated system in which the motherboard connects the components, the CPU processes instructions, RAM supports active tasks, the GPU handles graphics, storage saves data, the PSU provides power, and cooling fans help control temperature.